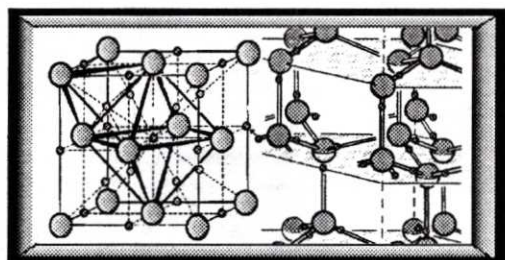


TO COLDLY GO WHERE NO ONE HAS GONE BEFORE



Metallic palladium (at left) is shown fully loaded with deuterium, an isotope of hydrogen. Heavy water (D_2O , on the right) is the source of the isotopic fuel used in some "cold fusion" reactions.

COLD FUSION TIMES

Volume 1 Number 3
FALL 1993

HEAVY WATERGATE & The Importance of Cold Fusion

Editorial: Cold fusion began as the generation of anomalous excess heat from an immersed palladium electrode, electrically activated, in heavy water. The technology has since blossomed to include anomalous heats of both steady-state and burst type, and has developed to include the use of titanium, nickel, and other metals. The reported excess heats are in the range of 2 to 400% for the steady state excess enthalpies and occasional heat bursts demonstrate even greater amounts of potential useful energy. Cold Fusion is important, even if there is only a 1/10 of 1% possibility of its reality, because it is inexpensive. Given that the fuel is contained in water, given its safety (consider helium versus acid rain or radioactive isotopes), and given its impact (jobs and improved welfare in the short-term horizon and a serious potential reduction of the balance-of-trade payments in the long-term), cold fusion must be investigated.

After controversy following their recent publication (Physics Letters A, 176 (1993), published May 3), Drs. Fleischmann and Pons answer their critics thereby correcting considerable misinformation.

Continued on page 3

from Valbonne, France

Response to Critique of Physics Letters A Paper

by Drs. Martin Fleischmann and Stanley Pons

We reply here to the critique by Douglas Morrison [1] of our paper [2] which was recently published. Apart from his general classification of our experiments into stages 1-5, we find that the comments made [1] are either irrelevant or inaccurate or both. In the article "Comments on Claims of Excess Enthalpy by Fleischmann and Pons using simple cells made to Boil" Douglas Morrison presents a critique [1] of the paper "Calorimetry of the Pd- D_2O system: from simplicity via complications to simplicity" which has recently been published in this Journal [2]. In the introduction to his critique, Douglas Morrison has divided the time-scale of the experiments we reported into 5 stages. In this reply, we will divide our comments into the same 5 parts. However, we note at the outset that Douglas Morrison has restricted his critique to those aspects of our own paper which are relevant to the generation of high levels of the specific excess enthalpy in Pd-cathodes polarized in D_2O solutions i.e. to stages 3-5. By omitting stages 1 and 2, Douglas Morrison has ignored one of the most important aspects of our paper and this, in turn, leads him to make several erroneous statements. We therefore start our reply by drawing attention to these omissions in Douglas Morrison's critique.

Continued on Page 2

from Japan

Cold Fusion in Japan

By Jed Rothwell, Cold Fusion Research Advocates

For people who read Japanese, it is easy to keep tabs on cold fusion research in Japan. Many technical journals publish papers. Most researchers are happy to talk informally about their work, and many have been interviewed in the popular press. Technical and scientific magazines equivalent to our Scientific American often carry articles about it, as do the major newspapers. At least five books have been published on the subject, including a translation of Peat's Cold Fusion, and one was written by Yamaguchi, a leading researcher. There is a surprising amount of English source material available too; journals published in English like the Japanese Journal of Applied Physics include many technical papers about cold fusion. However, as in any other field of science, business, industry, politics or culture, in order to get any real, in-depth information, the ability to read Japanese is essential, just as the ability to read English is essential for serious research in the U.S.

The Japanese approach to cold fusion is low key and pragmatic. It is impossible to know exactly how many

Continued on page 7

from Washington, D.C.

Report on an Important Meeting with the U.S. Congress

by J. Rothwell, E. Mallove

A "Cold Fusion Information Seminar" was held in the Rayburn House Office Building, July 23, 1993. Three representatives of Cold Fusion Research Advocates, Jed Rothwell (Director), Eugene F. Mallove (Associate Director), and Thomas Benson (Associate Director), organized a "Cold Fusion Information Seminar" for all members of Congress and their staff. The seminar included the half-hour video documentary, "The Secret Life of Cold Fusion", by the Canadian Broadcasting Corporation [Ed: see page 12]. The gathering took place in the U.S. House of Representatives Rayburn House Office Bldg., Room 2318, the usual hearing room of the House Science, Space, and Technology Committee, and where the ill-fated April 1989 hearings on cold fusion occurred! The CFRA

Continued on page 4

THE JOURNAL
DEVOTED TO
RELEVANT
RESEARCH and
DEVELOPMENTS
OF SAFE, RELIABLE
POWER SOURCES
USING HYDROGEN
ENERGY FROM A
MATERIAL

In March 1989
electrochemically
induced nuclear
fusion reactions
were reported, but
were initially very
difficult to
reproduce.

By mid 1993
many experiments
have demonstrated
that anomalous
nuclear processes
occur in palladium,
and other metals,
which have been
highly loaded with
deuterium.

This journal
reports on the
state-of-the-art
from around the
world, and in this
issue even beyond -

(see page 2 for
palladium mining in
the asteroid belt).

No Portion of this journal may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means, including electronic, print, photocopy, mechanical, recording or otherwise, without the prior permission of JET Technology c/o COLD FUSION TIMES

TABLE OF CONTENTS
ON PAGE 2

© copyright 1993
JET Technology

ISSN# 1072-2874
All rights Reserved

TABLE OF CONTENTS

CBC Video Tape		13
CF & the Future: Space Mining	D. Rotegard	2
CF Overview	G. Mallove	4
Cold Fusion in Japan	J. Rothwell	1
For the World		10
Heat-Off		3
Importance of Cold Fusion	M. Swartz	1
Remarks re: Physics Letters	M. Fleischmann, S. Pons	1
Report to U.S. Congress	J. Rothwell, G. Mallove	1
Short Excerpt	Sunday Times (London)	2
Tips on Nickel Systems	Mallove, Rothwell, Swartz	3
Trigger Survey		9
What's Happenin	various	11

BACK TOWARDS THE FUTURE

Ed. We thank Prof. Julius Mariasis for reminding us to direct this field forward. Economics will direct us this month. This excerpt is from *The Case for Mars III: Strategies for Exploration - Technical, Science and Technology Series, 75*, ed. Carol Stoker, 1989 [Univelt, CA]. The author, Dana Rotegard is an associate technical consultant for JANES SPACE MARKETS, and is an economist who herein teaches, amongst many things, the importance of mining the asteroid belt for palladium. As in the past, economics may drive the Earth's initial commitment of resources. The connection here to cold fusion is through there reserves of platinum group metals located in, or near, Mars. That and a dirty snowball from the Oort Cloud may enable Mankind to leave the Solar System.

from Minneapolis, MN

THE DEVELOPMENT OF SPACE: THE ECONOMIC CASE FOR MARS

Dana Richard Rotegard

I. SPACE INDUSTRIALIZATION

A. THE SATELLITE INDUSTRY In 1945, British author Arthur C. Clarke wrote a paper in *Wireless World*, where he advocated placing communications relays in geo-synchronous orbit around Earth, replacing ground systems and creating a worldwide telecommunications linkage. Clarke speculated that someone might launch such a satellite around the turn of the century. At the time, Clarke was widely ridiculed as a utopian dreamer. As of 1986, there were over 500 satellites in or planned in "Clarke Orbit." This satellite industry creates a \$500 million a year demand for launch services besides creating a demand for Earth stations and other satellite spinoffs. Plans for the further commercialization of space are afoot all over the world and, if experience with satellites is any indication, the development of near Earth orbit could take place faster than even its proponents imagined [Ed.: More on this in the last issue].

B. MATERIALS PROCESSING IN LEO As far back as the writings of Russian space pioneer Konstantine Tsiolkovsky in the 19th century, visionaries have speculated about manufacturing in space. Hard vacuum and microgravity create possibilities for materials processing that are impossible or prohibitively expensive

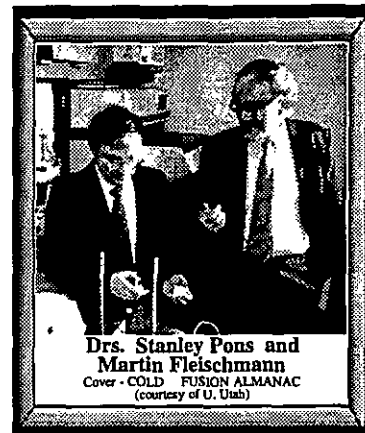
Continued on page 5

Response from Drs. Fleischmann and Pons continued from Page 1

Stages 1 and 2 In the initial stage of these experiments the electrodes (0.2mm diameter x 12.5mm Pd-cathodes) were first polarized at 0.2A, the current being raised to 0.5A in stage 2 of the experiments. We note at the outset that Douglas Morrison has not drawn attention to the all important "blank experiments" illustrated in Figs 4 and 6 of our paper by the example of a Pt cathode polarized in the identical 0.1M LiOD electrolyte. By ignoring this part of the paper he has failed to understand that one can obtain a precise calibration of the cells (relative standard deviation 0.17%) in a simple way using what we have termed the "lower bound heat transfer coefficient, $(k_r)_{11}$ ", based on the assumption that there is zero excess enthalpy generation in such "blank cells". We have shown that the accuracy of this value is within 1 sigma of the precision of the true value of the heat transfer coefficient, $(k_r)_2$, obtained by a simple independent calibration using a resistive Joule heater. Further methods of analysis [3] (beyond the scope of the particular paper [2]) show that the precision of $(k_r)_{11}$ is also close to the accuracy of this heat transfer coefficient (see our discussion of stage 3). We draw attention to the fact that the time-dependence of $(k_r)_{11}$, (the simplest possible way of characterizing the cells) when applied to measurements for Pd-cathodes polarized in D_2O solutions, gives direct evidence for the generation of excess enthalpy in these systems. It is quite unnecessary to use complicated methods of data analysis to demonstrate this fact in a semi-quantitative fashion.

Stage 3 Calculations Douglas Morrison starts by asserting: "Firstly, a complicated non-linear regression analysis is employed to allow a claim of excess enthalpy to be made". He has failed to observe that we *manifestly have not used this technique in this paper* [2], the aim of which has been to show that the simplest methods of data analysis are quite sufficient to demonstrate the excess enthalpy generation. The only point at which we made reference to the use of non-linear regression fitting (a technique which we used in our early work [4]) was in the section dealing with the accuracy of the lower bound heat transfer coefficient, $(k_r)_{11}$, determined for "blank experiments" using Pt-cathodes polarized in D_2O solutions. At that point we stated that the accuracy of the determination of the coefficient $(k_r)_2$ (relative standard deviation ~1.4% for the example illustrated [2]), can be improved so as to be better than the precision of $(k_r)_{11}$ by using non-linear regression fitting; we have designated the values of (k_r) determined by non-linear regression fitting by $(k_r)_3$. The values of $(k_r)_3$ obtained show that the precision of the lower bound heat transfer coefficient $(k_r)_{11}$ for "blank experiments" can indeed be taken as a measure of the accuracy of $(k_r)_3$. For the particular example illustrated the relative standard deviation was ca. 0.17% of the mean. It follows that the calibration of the cells using such simple means can be expected to give calorimetric data having an accuracy set by this relative standard deviation in the subsequent application of these cells.

We note here that we introduced the particular method of non-linear regression fitting (of the numerical integral of the differential equation representing the model of the calorimeter to the experimental data) for three reasons: firstly, because we believe that it is the most accurate single method (experience in the field of chemical kinetics teaches us that this is the case); secondly, because it avoids introducing any personal bias in the data treatment; thirdly, because it tends to direct estimates of the standard deviations of all the derived values from the diagonal elements of the error matrix. However, our experience in the intervening years has shown us that the use of



Drs. Stanley Pons and
Martin Fleischmann
Cover - COLD FUSION ALMANAC
(courtesy of U. Utah)

from London, U.K.

Nuclear confusion

Short excerpts from *The Sunday Times* (27 June 1993) by Neville Hodgkinson

"... according to Ala Bewick, other British scientists were eager to jump on to the anti-fusion bandwagon. 'Of all the countries I know, the UK is the most negative towards cold fusion,' he says. "A large number of scientists, not only physicists, are incredibly emotional and personal about it. "Britain could have been in the lead on this. Martin was keen to have things done in Britain, but in the end the physics establishment went totally against. You can see why. Physicists think they have an awful lot to lose from this. They think, 'Mere chemists, dealing with something that looks like a nuclear phenomenon...'"

*** "Meanwhile, three international conferences on cold fusion have taken place, but have been regarded by most Western scientists simply as conventions for "believers". The latest took place last October at Nagoya in Japan. Of 320 scientists assembled there, 199 came from Japan itself, 55 from the US, 20 from Italy, 11 from China, five from France, two from Germany and only one from the UK-Fleischmann."

"The "believers" have put forward theories that might underpin the phenomenon, and have continued to report a variety of anomalous experimental results, many appearing to support the original claims. But a lack of consistency in the data has allowed sceptics to maintain that experimental errors are a more likely explanation than newly discovered physical laws."

"As far as, 'official', science was concerned, the cold fusion episode became regarded as an unfortunate incident best forgotten. And there the matter seemed to rest - until now."

"The past few months have seen developments which, according to Pons, are making even hardened sceptics soften their stance. "It's moving forward," he says. "The fact that you get excess heat has been pretty much verified. It has been borne out by hundreds of papers and experiments. Our own results are consistent in this, and the thermal outputs we now measure are about 100 times what we saw in early 1989."

"The cold fusion story raises some major questions about the scientific establishment's openness to new ideas, and industry's willingness to fund research into them. According to Eiichi Yamaguchi, senior NTT scientist, Britain and America decided to take the easy way out of the cold fusion furore. "They don't like confusion," he says. "They don't like basic research where only one in 10,000 experiments is successful."

Because of rapidly developing technologies in cold fusion research, COLD FUSION TIMES is available for home or office delivery. COLD FUSION TIMES brings up-to-date, relevant information to its readers, and an open forum for all views on cold fusion. The COLD FUSION TIMES is not responsible for the comments of its reporters or readers whose words remain protected by the First Amendment of the U.S. Constitution. Questions, comments, suggestions, and contributors are welcome.

For editorial information, please call Mitchell Swartz
(617) 239-8383 or Internet: mica@world.std.com.
P.O. Box 81135
Wellesley Hills, MA 02181

© copyright 1993
JET Technology

COLD FUSION TIMES®
is a trademark of
JET Technology

COLD FUSION TIMES

Editorial : Continued from page 1

The knowledge expands elsewhere as well. Where (and why) has there been such a distribution of effort? So little in the United States and the United Kingdom? And the drift of the main players (Messieurs F&P) to France. Have we seen this before? Earlier American inventors had sold their products to France and Germany, even while both the U.S. military and the U.S. Patent Office ignored the marvelous inventions. That one of these inventions was the airplane should be most thought-provoking, but it is also quite intriguing because France had not always acted this clever way. Emp. Napoleon turned down the American invention of the steam powered boat. Napoleon guessed wrongly that a raging fire -- presumably contained below deck -- could never be used to power any boat against the wind. France apparently learned well from that error.

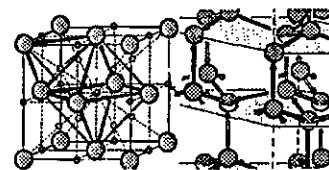
It is ironic that the core of "cold fusion-skeptics" are led by "hot fusion" and other physicists who have unfortunately failed to demonstrate even one minute of excess heat on their hot fusion systems. There remains not one "working" continuous hot fusion system despite billions of dollars over decades. In contrast, cold fusion is doing quite well on bootstrapped funds. Imagine if support had been given earlier to those investigators who have pursued it. Drs. Fleischmann and Pons demonstrated a novel system demonstrating unexpected physics in March 1989. Then, the reported excess power densities were about 20 Watts per cubic centimeter of palladium. Today, the amount of excess heat capable of being produced has actually increased by a factor of 10,000% based upon that formulation!

In rare public sharing of information, partly as a result of provocation by the skeptics, and in part by promethean barriers to publication, and tragically secondary to a series of explosions (one of which was lethal), cold fusioners have attempted to come together to discuss cold fusion and to clear the air. The clearing process may yet be shown to be a **HEAVY WATERGATE**.

Drs. Martin Fleischmann and Stanley Pons may have initially underemphasized that the reactions are very difficult-to-prepare. Much early effort was upon the use of alloys, additives, and other techniques. One quantum advance occurred when Dr. Michael McKubre (SRI; see **COLD FUSION TIMES** issue #1) demonstrated that the addition of aluminum after partial loading of the palladium by deuterium appears to increase the likelihood of generating these reactions. Dr. Edmund Storms clarified that not all palladium electrodes are equal (see **COLD FUSION TIMES** issue #2). Why is this yet another quantum advance? Because some electrodes work and some do not; and wisdom in this field will consist of knowing the difference. Dr. Melvin Miles (China Lake, US Navy; see issue #2) has demonstrated that helium, used to fill ordinary lighter-than-air balloons, is one product cold fusion, and it is generated by, and linked to, the production of excess heat.

Why is there so much apparent anti-cold fusion sentiment? Several misstatements ought to be challenged because destructive criticism is just too easy to make. The skeptics claim that cold fusion is not reproducible. However, they obviously have NOT attempted to reproduce either the experiments of Dr. Miles, Dr. Storms, Dr. McKubre, or others. Furthermore, many papers they quote do not describe experiments performed correctly.

The skeptics' oft-quoted "negative papers" have been shown to contain major errors, and most of the experiments described in the papers have failed to achieve adequate loading of the palladium. Some used poor calibration, another removed the steady-state signal, and at least one may have failed to report a functioning cell.



HEAT-OFF

The **COLD FUSION TIMES** is accepting suggestions for a **HEAT-OFF** contest to be held on the Fifth Anniversary of the **COLD FUSION** announcement. **JET Technology** and **FEAT** (through the courtesy of Hal Fox) will offer an award to the best low power system.

To speed this along, we present the first in a series of hints regarding the cold fusion phenomena



graphic after Fortean Magazine
Issue 69, July 1993

Suggestions for Conducting Light Water Excess Energy Experiments

Jed Rothwell, Eugene F. Mallove, M. Swartz

Although there have been explosions and other dangers associated with the deuterium/palladium systems, there does not appear to be that danger with the protium (that is, light water) nickel systems. Many people have heard of the light water excess energy experiment reported by Mills and Kneizys in *Fusion Technology*. By January, 1992, this excess energy effect had been reproduced by at least a half-dozen other groups. Even though the experiment is simple and apparently highly reproducible, many would-be experimenters might be deterred from trying it because of the well-known history of difficulties with the heavy water palladium-platinum approach of Fleischmann and Pons. Even though Mills et alia do not think that their excess energy is due to "cold fusion" -- they have an elaborate theory of shrinking hydrogen atoms to explain the excess power -- their experiment was inspired by the Fleischmann-Pons announcement. The purpose of this brief collection of experimental protocols will be to encourage others to try the Mills experiment and perhaps go beyond it in their investigations. Beginning next month we will discuss the systems, the materials, and some of the tips. However, at the time, the first order of business is to read the experimental part of the Mills-Kneizys paper in *Fusion Technology* to familiarize yourself with the basic approach. [Mills, Randall L. and Steven P. Kneizys, "Excess Heat Production by the Electrolysis of an Aqueous Potassium Carbonate Electrolyte and the Implications for Cold Fusion," *Fusion Technology*, Vol.20, August 1991, pp.65-81. Also: Noninski, V.C., "Excess Heat During the Electrolysis of a Light Water Solution of K_2CO_3 With a Nickel Cathode," *Fusion Technology*, March 1992.]

The skeptics incorrectly claim that the "light water work disproves "cold fusion". In fact, the light water experiments occur on a totally different metal; and that is nickel and not palladium. An actual reading of the literature would clarify this.

The skeptics have claimed that hundreds of million dollars have been spent on cold fusion each year and that the taxpayers fund this. This is also not correct. To the contrary, there is minimal to no money within the continental United States for this work (except on an underground or ad hoc basis) and that is why the majority of the research and development is taking place **TODAY** in the Japan, France, Russia, India, China, and Italy.

The skeptics have claimed that nonreproducible events are not science. That is not true. Earthquakes are hardly reproducible but geologists struggle and continue with their efforts to help humankind. Can you imagine stopping the study of Earthquakes (or asteroid hits)?

The skeptics have claimed that all of the products of cold fusion are, or must be, radioactive. This also is not true. One product: helium is in the very air we breathe, and is simply not radioactive, but is rather a

harmless gas.

The skeptics have claimed that the products generated ["ash"] do not exactly match the heat generated. But the products are difficult to measure and are produced in very small amounts because of $E=mc^2$. Some skeptics even demand that Mother Nature produce pathways which match up to their predictions predicated upon hot (plasma) fusion.

The skeptics claim there should be "no revaluation of cold fusion". However, the "body" of cold fusion actually has a strong pulse, and is warming up. An injection of funds, and improvement of image, will go far to help correct the situation. Those with curiosity will continue to explore these phenomena. Those with the inner desire to help humanity will innovate the field.

Best wishes.

Mitchell Swartz
Editor, **COLD FUSION TIMES**
Internet mica@world.std.com

U.S. Congressional Hearing

CONTINUED FROM PAGE 1

distributed additional copies of the video, the August Popular Science magazine, copies of the Clarke PASOLS speech, and technical papers by Pons and Fleischmann, McKubre, Miles and others.

Members of Congress and their staff, whether or not they attended the CBC video showing, now have in their hands all the information they need to determine that cold fusion is an active field of investigation in Japan, in the U.S., and elsewhere. After the CBC video was shown for the first time, Jed Rothwell read the following statement to the assembled guests:

"My area of expertise is cold fusion in Japan. The video touched on this subject, but it did not bring out the fact that Japan leads the world in this research. I speak Japanese, and I have been in contact with researchers, government officials, and science journalists there frequently. The information I am presenting today comes from national newspapers, technical magazines and journals, and Official Japanese Government publications. Let me address a few questions that people often ask me.

Question 1: What is the Ministry of International Trade and Industry (MITI) doing? In 1992, MITI organized a cold fusion research and development consortium including 10 of Japan's largest corporations. In the Spring of 1992, the newspapers reported that five other corporations asked to join. Public interest in cold fusion picked up, and last July MITI's plans were featured in the mass media. The Yomiuri newspaper trumpeted the news in a first page headline: "Nation Begins Full Scale Research." Cold fusion research in Japan is not secret. Last fall, MITI invited me to a planning session of the consortium, where I learned about their plans in some detail. The members of that consortium are listed on the back of my Fact Sheet handout. They include: the Chubu Electric Power, Hitachi, Toshiba, Fuji Electric, Toyota and others. The research is directed by a team of top scientists from these corporations, lead by the distinguished Professors Pons and Fleischmann, in their role as senior advisors to MITI. They work for Toyota, but they also help direct the government program.

Question 2: What is the size of the effort? I estimate government and private industry in Japan together spend at least \$90 million per year on cold fusion. The number that you hear bandied about sometimes is \$25 million over 4 years from MITI, but that is only the tip of the iceberg. The director of the MITI program explained that the \$25 million is a special fund for extra equipment, travel, and other expenses at the National Laboratories which exceed normal discretionary funding levels. To get an idea of how serious they are, let me point to the handout again. The list of Official Sponsors for the Nagoya Conference includes: The Physical Society of Japan, The Japan Society of Applied Physics, Atomic Energy Society of Japan, The Institute of Electrical Engineers of Japan. Here is the latest edition of a top scientific publication: the Journal of the Japan Society of Applied Physics. It includes 6 articles on cold fusion. Forty-six Japanese corporations sent representatives to the Nagoya Conference; they are listed in the handout. The head of the Government program, Ikegami, estimates that 20 of the big corporations are "serious about this research." Some of the smaller companies I talked to had only one or two people. Big companies appeared to have 10 or 20, and Toyota, which is the biggest, has at least 80 by my count. Very roughly, I estimate there are 150 scientists in government and at least 450 in industry, a total of 600. With salary and overhead it comes to \$90 million dollars. That does not count capital expenditures like the Toyota lab you saw in the video. It was built from the ground up for that purpose, so it must have cost tens of millions.

Question 3: How close are they to commercialization? The video gave you a good idea of that. Starting from scratch in 1989, they have increased the strength of the reaction 100 fold. They achieved 150 watt reactions last year, with 10 times the power density of commercial fission reactor rods, and high temperatures which allow boiling water. This is a key point: if you want to generate electricity, you have to boil water. By the end of this year, as you saw, they hope to have 10 to 20,000 watt reactors that operate continuously. MITI has said that they expect to see prototype electric power generators by 1997. They have achieved 100% reproducibility on demand. There are no signs of any roadblock, no shadow of any future technical limitation, and no unsolved problems have been left behind to be dealt with "later." So, I believe MITI's prediction: prototype power reactors by 1997.

The last question: **What do they think it is?** Why have they embraced it? Well, they are not sure whether it is nuclear fusion or not. They know it is not an experimental error, and they know it is not chemical energy. So, every single Japanese scientist and government official has told me: "We don't know what it is. That's what we are trying to find out." These are pragmatic people. In 1989, they decided that cold fusion might just be a real phenomenon, so they decided to investigate it. They found out it works. As the Director of the MITI program said to me, "I don't care what it is, as long as it works."

Following Jed's statement, Dr. Eugene Mallove quoted Professor Ikegami from the Proceedings of the Nagoya Conference: "It is my belief that cold fusion will become one of the most important subjects in science, one for which we have been working so patiently, with dedication and with courage,

Continued on page 8

from Bow, New Hampshire

Overview of COLD FUSION

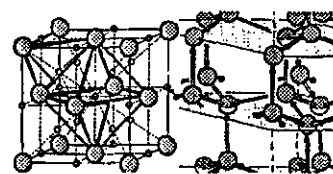
by Eugene Mallove, J. Rothwell

In 1993, unfortunately, a great deal of the cold fusion story still centers around political conflict, controversy, and raw emotions: hate, jealousy, rivalry, animosity, and fear of the unknown. The "skeptics" are still fighting a rear guard delaying action; they must deny there is any validity to the field, because if they admit that even a single experiment is valid, they open the floodgates. Positive cold fusion experiments have now been performed over and over again in hundreds of top-notch laboratories worldwide, ranging from Los Alamos National Laboratory, SRI International and Texas A&M University, to BARC (Bhabha Atomic Research Center) India; from Osaka University, Tokyo Institute of Technology, to Toyota. The "skeptic" is pushed into an unenviable position: he or she must insist that every single one of these experiments is utterly wrong. To admit that even one excess heat measurement - or one tritium measurement - is right, is to admit that a nuclear effect exists in metals, where by conventional physics it has absolutely no "right" to be. Nuclear reactions are not supposed to be initiated in chemical systems near room temperature, period! So all claims must be denied, absolutely, no matter how convincing the experiments. Do you suppose Toyota could confuse a temperature of 45 deg C with 100 deg C, or even 300 deg C? Do you suppose three independent teams of veteran safety experts and other scientists at India's best nuclear facility could mistake zero tritium for tritium at levels hundreds of times higher than normal, in dozens of experiments? If you are a "skeptic" who wishes to deny every scrap of data, you must defend these points of view. Your case must be airtight, because if even one experiment is true then not only is the effect real, but after all, why shouldn't the other experiments also be true?

So, unfortunately, cold fusion in the U.S. remains mired in irrational denials and hysterical opposition. It is a bleak short-term picture, but the reader must not allow this to cloud the essentially positive, hopeful future outlook. There can be no question now that enough momentum has developed in Japan to carry cold fusion development through to an inevitable, positive conclusion. Dozens of Japan's top corporations are working in the field, and a whole array of Japan's leading organizations like the Physical Society, the Society of Applied Physics, the Atomic Energy Society of Japan, and the IEEE regularly sponsor important meetings and conferences. Japan's Trigger Magazine conducted a survey in May, 1993, asking 300 leading industrialist and scientists what they thought of cold fusion. 190 responded (a phenomenally high 69% response rate). Of these, 95% said they were "very interested." When asked why cold fusion experiments are still so difficult to do, 27 people expressed the typical U.S. "skeptical" point of view that it is because the phenomenon does not exist; 23 said it is because the experiments have not been done very well, so the results have not been clear cut; 127 said it is simply because this is new science, and it takes time to learn how to do this kind of thing; and 3 said they were not sure.

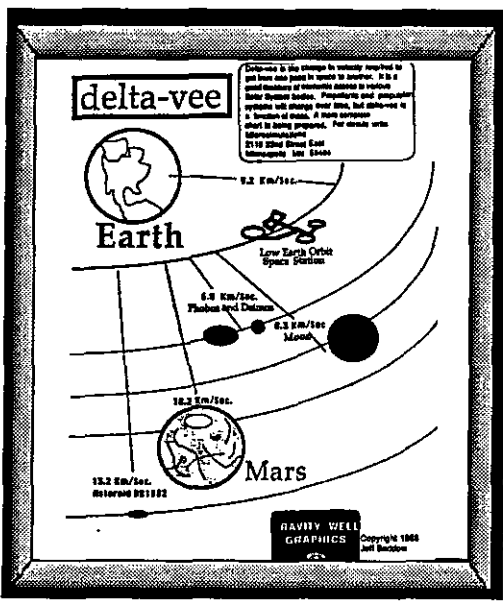
The level of interest and knowledge of cold fusion in the U.S. and Europe is far smaller than in Japan! Most business and academic leaders think that cold fusion disappeared in 1989. However, it is important to note that despite the U.S. government's appalling treatment of cold fusion, private research institutions in the U.S. are carrying the ball for the science. Cold fusion will almost certainly become a useful form of energy, in fact, we think it is likely to become the overwhelmingly dominant form of energy for the foreseeable future. However, this outcome cannot be perfectly certain - nothing in life can be. No matter how good the prospects for a technology might appear, only a fool counts his chickens before they hatch. Cold fusion looks very promising at the moment. There appear to be no roadblocks and no reason to doubt that power reactors and other practical sources of energy will appear within a few years. However, in the annals of the history of technology, there are examples of last minute, unforeseen problems cropping up and derailing inventions. Freeman Dyson gives a good example in his book Weapons and Hope: the Automatic Gun Laying Turret (AGLT), which was a radar guided anti-aircraft gun mounted on board British bombers in 1944. It could shoot down attacking German fighter planes at night even more effectively than a conventional, manually operated machine gun could during the day. The only problem with it was that given the actual conditions of air warfare, it was found to be far more likely to shoot down another British bomber than a German fighter. The IFF (Identify Friend or Foe) systems in 1944, that should have allowed the gun to automatically discriminate between British and German airplanes were only about 90% effective, so if the AGLT guns had been used as planned, they would have shot down 40 British airplanes for each German. The IFF could not be improved sufficiently, so the AGLT could only be used

Continued on page 5



COLD FUSION TIMES

THE DEVELOPMENT OF SPACE
CONTINUED FROM PAGE 2



The case for mining platinum group metals (PGM) from asteroids is based upon the scarcity of these materials on Earth.

left) Δ-vee is the change in velocity required to get from one point in space to another. This graph is from Jeff Beddow of Gravity Well Graphics and is discussed in Dr. Rotegard's article.

OVERVIEW of COLD FUSION
CONTINUED FROM PAGE 4:

after visual contact was established, which meant it was only marginally better than a manually operated gun.

It is hard to imagine how cold fusion might run into a similar problem, but of course, it the very nature of such problems is that they cannot be anticipated! Other forms of nuclear energy started out looking very promising in the 1940's but later proved to be difficult to perfect, and economically uncompetitive. It is conceivable - but we think highly unlikely - that cold fusion could remain a fascinating but impractical laboratory curiosity, perhaps for decades, perhaps even forever. It is however, very difficult to imagine how cold fusion might be developed but then found to be somewhat dangerous and economically uncompetitive like fission energy. Cold fusion appears to be so simple, and so safe, that if it works at all it will be an instant triumph in all areas. It will be cost effective, economical, safe, and an environmental dream come true. Certainly, from a technical standpoint, cold fusion has far fewer drawbacks and potential problems than fission did in 1945. There is nothing toxic, and nothing the least bit dangerous about a cold fusion reactor; it appears to be far safer than chemical combustion - fire. It does not appear to have any ill effects on any form of life. Naturally, before cold fusion generators can be mass produced, this will have to be confirmed by extensive testing with laboratory animals (and even plants). Potentially harmful radiations that can be detected have been measured only a very low levels, and even these give every sign of being "tunable" to even lower levels. The main problem at the moment is that there is no generally accepted theory of the operation of cold fusion, so we do not know exactly how to "tune" them yet.

It is worth remembering that when new phenomena have been discovered in the past which did turn out to have unexpected ill effects, this fact was discovered very quickly, often within months of the invention. For example, soon after Roentgen announced the discovery of X-rays in 1895, doctors all over the world were using X-ray machines to assist in diagnosis and surgery. Within months, they overdid it while looking for a bullet lodged in the body of a patient in New York City. The patient died gruesomely, and the doctors operating the machine also suffered terrible effects, and everyone knew at once that the X-ray machine had to be used with caution. Soon after the discovery of radioactivity in 1896, this, too was found to be dangerous, although not in time to save Madam Curie from an early death from cancer. If years of direct exposure to cold fusion has any ill effects, we shall soon find out about them through careful laboratory experiments with rats and other animals, and also because cold fusion scientists themselves are acting as unwitting "guinea pigs" - they are exposed day and day out to unshielded devices. If it turns out that cold fusion cells do cause some damage, or they do liberate tritium at a level say, one-thousand times lower than the natural radioactive background, then steps may have to be taken to contain these threats. Perhaps we could design cold fusion reactors to shut down automatically if tritium production suddenly begins, or some other threat materializes. This would be no different in principle from building things like self-sealing fuel tanks on airplanes, and fuses in houses to prevent fires. Threats can be contained and neutralized to a reasonable, livable level. Cold fusion will be far easier to control and make safe than oil or fission power.

on Earth.

C. LAUNCH TECHNOLOGY Most of the planning for space manufacturing is built around the capabilities of existing and proposed launch systems. The ability to supply raw materials, change crews, and return payloads are essential to this effort. The Soviets have recently added a large launcher (Energia) and a yet unlaunched shuttle to their existing family of launch systems and clearly have the best access to Low Earth Orbit for commercial purposes. Jane's Spaceflight reports that plans are afoot in Britain, France, Germany, and Japan to develop reusable manned orbital vehicles. Britain has the Hotel vehicle, a spacecraft similar in concept to America's Tokyo Express. The patent on the Hotel engine is held by Alan Bond and engine development is licensed to Rolls Royce with British Aerospace to build the aircraft. The French are developing Hermes in conjunction with their Ariane launches. Japan's NASDA also has a rocket launched spaceplane in planning. The Federal Republic of Germany has its Sanger project.

The implications are clear. There is a worldwide effort to create new launch vehicles and exploit the commercial potential of LEO. This commercial explosion should be in full swing by the mid-1990's with most of the major industrial space-faring powers competing for markets in the space manufacturing area. Jerry Grey of AEROSPACE AMERICA estimates that by the year 2000 there will be a total market of 10-20-billion dollars for LEO manufacturing. Like Clarke's communications satellites before, space manufacturing promises great rewards to corporations and nations that can master the technology and exploit the opportunities. AEROSPACE AMERICA further estimates that by the year 2000 satellite communications, LEO manufacturing and their related launch services will be the core components of a 100 billion dollar a year near-earth space industry.

II. TRAJECTORY RESEARCH The growing manned and unmanned industrial complex in near-earth space in the 1990's will have to be supplied. Major options to supply these bases include supplies from Earth, the moon, the Mars system, or Earth-crossing asteroids. In the early days of the German rocket clubs and the British Interplanetary Society visionaries in the field of interplanetary rocket travel realized that distance as measured in kilometers through a near vacuum had very little relation to the engineering difficulty or expense of various deep space missions. Delta V, or the total change in velocity needed to propel a spacecraft from one orbit to another in the solar

Continued page 8



Cold Fusion Impact
by Hal Fox, Ed. Fusion Facts
(see Page 10)



Cold Fusion Almanac
by Cold Fusion Research Associates
(see Page 10)

Response from Drs. Fleischmann and Pons

Continued from page 2

this method is a case of "overkill": it is perfectly sufficient to use simpler methods such as multi-linear regression fitting if one aims for high accuracy. This is a topic which we will discuss elsewhere [3]. For the present, we point out again that the purpose of our recent paper [2] was to illustrate that the simplest possible techniques can be used to illustrate the generation of excess enthalpy. It was for this reason that we chose the title: "Calorimetry of the Pd-D₂O system: from simplicity via complications to simplicity".

Douglas Morrison ignores such considerations because his purpose evidently is to introduce a critique of our work which has been published by the group at General Electric [5]. We will show below that this critique is totally irrelevant to the recent paper published in this Journal [2]. However, as Douglas Morrison has raised the question of the critique published by General Electric, we would like to point out once again that we have no dispute regarding the particular method of data analysis favoured by that group [5]: their analysis is in fact based on the heat transfer coefficient (k_h)₂. If there was an area of dispute, then this was due solely to the fact that Wilson et al introduced a subtraction of an energy term which had already been allowed for in our own data analysis, i.e. they made a "double subtraction error". By doing this they derived heat transfer coefficients which showed that the cells were operating endothermically, i.e. as refrigerators! Needless to say, such a situation contravenes the Second Law of Thermodynamics as the entropy changes have already been taken into account by using the thermoneutral potential of the cells. We will leave others to judge whether our reply [6] to the critique by the group at General Electric [5] did or did not "address the main questions posed by Wilson et al." (in the words of Douglas Morrison). However, as we have noted above the critique produced by Wilson et al [5] is in any event irrelevant to the evaluations presented in our paper in this journal [2]: we have used the self-same method advocated by that group to derive the values of the excess enthalpy given in our paper. We therefore come to a most important question: "given that Douglas Morrison accepts the methods advocated by the group at General Electric and, given that we have used the same methods in the recent publication [2] should he not have accepted the validity of the derived values?"

Stage 4 Calculation Douglas Morrison first of all raises the question whether parts of the cell contents may have been expelled as droplets during the later stages of intense heating. This is readily answered by titrating the residual cell contents: based on our earlier work about 95% of the residual lithium deuteride is recovered; some is undoubtedly lost in the reaction of this "aggressive" species with the glass components to form residues which cannot be titrated. Furthermore, we have found that the total amounts of D₂O added to the cells (in some cases over periods of several months) correspond precisely to the amounts predicted to be evolved by (a) evaporation of D₂O at the instantaneous atmospheric pressures and (b) by electrolysis of D₂O to form D₂ and O₂ at the appropriate currents; this balance can be maintained even at temperatures in excess of 90 degrees C [7].

We note here that other research groups (eg [5]) have reported that some Li can be detected outside the cell using atomic absorption spectroscopy. This analytic technique is so sensitive that it will undoubtedly detect the expulsion of small quantities of electrolyte in the vapour stream. We also draw attention to the fact that D₂O bought from many suppliers contains surfactants. These are added to facilitate the filling of NMR sample tubes and are difficult (probably impossible) to remove by normal methods of purification. There will undoubtedly be excessive foaming (and expulsion of foam from the cells) if D₂O from such sources is used. We recommend the routine screening of the sources of D₂O and of the cell contents using NMR techniques. The primary reason for such routine screening is to check on the H₂O content of the electrolytes.

Secondly, Douglas Morrison raises the question of the influence of A.C. components of the current, an issue which has been referred to before and which we have previously answered [4]. It appears that Douglas Morrison does not appreciate the primary physics of power dissipation from a constant current source controlled by negative feedback. Our methodology is exactly the same as that which we have described previously [4]; it should be noted in addition that we have always taken special steps to prevent oscillations in the galvanostats. As the cell voltages are measured using fast sample-and-hold systems, the product ($E_{\text{cell}} - E_{\text{thermoneutral, base}}$) will give the mean enthalpy input to the cells: the A.C. component is therefore determined by the ripple content of the current which is 0.04%.

In his third point on this section, Douglas Morrison appears to be re-establishing the transition from nucleate to film boiling based on his experience of the use of bubble chambers. This transition is a well-understood phenomenon in the field of heat transfer engineering. A careful reading of our paper [2] will show that we have addressed this question and that we have pointed out that the transition from nucleate to film boiling can be extended to 1-10kW cm⁻² in the presence of electrolytic gas evolution.

Fourthly and for good measure, Douglas Morrison once again introduces the question of the effect of a putative catalytic recombination of oxygen and deuterium (notwithstanding the fact that this has repeatedly been shown to be absent). We refer to this question in the next section; here we note that the maximum conceivable total rate of heat generation ~5mW for the electrode dimensions used) will be reduced because intense D₂ evolution and D₂O evaporation degasses the oxygen from the solution in the vicinity of the cathode; furthermore, D₂ cannot be oxidised at the oxide coated Pt-anode. We note furthermore that the maximum localised effect will be observed when the density of the putative "hot spots" will be 1/δ² where δ is the thickness of the boundary layer. This gives us a maximum localised rate of heating of ~6nW. The effects of such localised hot spots will be negligible because the flow of heat in the metal (and the

solution) is governed by Laplace's Equation (here Fourier's Law). The spherical symmetry of the field ensures that the temperature perturbations are eliminated (compare the elimination of the electrical contact resistance of two plates touching at a small number of points). We believe that the onus is on Douglas Morrison to devise models which would have to be taken seriously and which are capable of being subjected to quantitative analysis. Statements of the kind which he has made belong to the category of "arm waving".

Stage 5 Effects In this section we are given a good illustration of Douglas Morrison's selective and biased reporting. His description of this stage of the experiments starts with an incomplete quotation of a single sentence in our paper. The full sentence reads: "We also draw attention to some further important features: provided satisfactory electrode materials are used, the reproducibility of the experiments is high; following the boiling to dryness and the open-circuiting of the cells, the cells nevertheless remain at a high temperature for prolonged periods of time (fig 11); furthermore the Kel-F supports of the electrodes at the base of the cells melt so that the local temperature must exceed 300 degrees C". Douglas Morrison translates this to: "Following boiling to dryness and the open-circuiting of the cells, the cells nevertheless remain at high temperature for prolonged periods of time; furthermore the Kel-F supports of the electrodes at the base of the cells melt so that the local temperature must exceed 300 degrees C".

Readers will observe that the most important part of the sentence, which we have underlined, is omitted; we have emphasized the words "satisfactory electrode materials" because that is the nub of the problem. In common with the experience of other research groups, we have had numerous experiments in which we have observed zero excess enthalpy generation. The major cause appears to be the cracking of the electrodes, a phenomenon which we will discuss elsewhere.

With respect to his own quotation Douglas Morrison goes on to say: "No explanation is given and fig 10 is marked 'cell remains hot, excess heat unknown'". The reason why we refrained from speculation about the phenomena at this stage of the work is precisely because explanations are just that: speculations. Much further work is required before the effects referred to can be explained in a quantitative fashion. Douglas Morrison has no such inhibitions, we believe mainly because in the lengthy section Stage 5 Effects he wishes to disinter "the cigarette lighter effect". This phenomenon (the combustion of hydrogen stored in palladium when this is exposed to the atmosphere) was first proposed by Kreysa et al [8] to explain one of our early observations: the vapourisation of a large quantity of D₂O (~500ml) by a 1cm cube palladium cathode followed by the melting of the cathode and parts of the cell components and destruction of a section of the fume cupboard housing the experiment [9]. Douglas Morrison (in common with other critics of "Cold Fusion") is much attached to such "Chemical Explanations" of the "Cold Fusion" phenomena. As this particular explanation has been raised by Douglas Morrison, we examine it here.

In the first place we note that the explanation of Kreysa et al [8] could not possibly have applied to the experiment in question: the vapourisation of the D₂O alone would have required ~1.1MJ of energy whereas the combustion of all the D in the palladium would at most have produced ~8.4 KJ (assuming that the D/Pd ratio had reached ~1 in the cathode), a discrepancy of a factor of ~130. In the second place, the timescale of the explanation is impossible: the diffusional relaxation time is ~29 days whereas the phenomenon took at most ~6 hours (we have based this diffusional relaxation time on the value of the diffusion coefficient in the alpha-phase; the processes of phase transformation coupled to diffusion are much slower in the fully formed Pd-D system with a corresponding increase of the diffusional relaxation time for the removal of D from the lattice). Thirdly, Kreysa et al [8] confused the notion of power (Watts) with that of energy (Joules) which is again an error which has been promulgated by critics seeking "Chemical Explanations" of "Cold Fusion". Thus Douglas Morrison reiterates the notion of heat flow, no doubt in order to seek an explanation of the high levels of excess enthalpy during Stage 4 of the experiments. We observe that at a heat flow of 144.5W (corresponding to the rate of excess enthalpy generation in the experiment discussed in our paper [2]) the total combustion of all the D in the cathode would be completed in ~4.5 s, not the 600s of the duration of this stage. Needless to say, the D in the lattice could not reach the surface in that time (the diffusional relaxation time is 10³ s) while the rate of diffusion of oxygen through the boundary layer could lead at most to a rate of generation of excess enthalpy of ~5mW.

Douglas Morrison next asserts that no evidence has been presented in the paper

Continued page 8

BE A SUPPORTER

Please renew your subscription rather than photocopy.
Thank you.

We are preparing the next issue of this journal which now has readership around the World. We want to invite you like to submit a contribution, an announcement, and your product information in this field. The COLD FUSION TIMES is the ONLY journal solely involving cold fusion and its derivative disciplines.

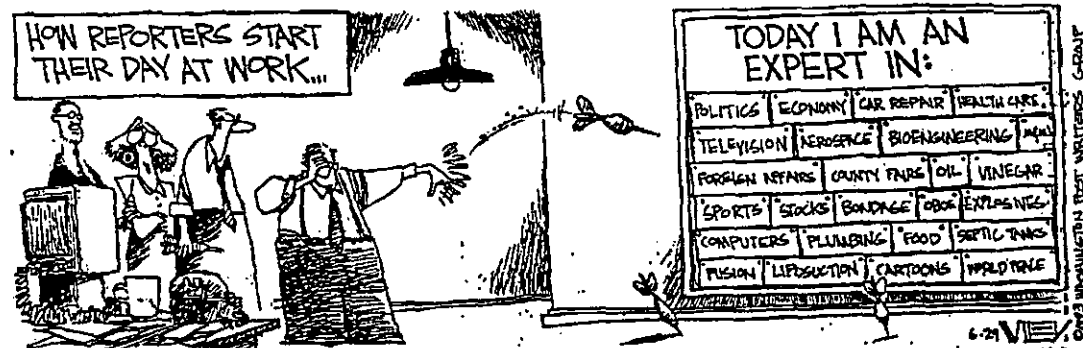
COLD FUSION TIMES

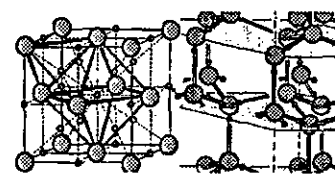
P.O. Box 81135
Wellesley Hills, MA 02181
e-mail: mcca@world.std.com

Our Favorite Cartoon of the Quarter

Ed: "Even Fusion?"
Boston Globe

NON SEQUITUR by Wiley





COLD FUSION TIMES

Cold Fusion in Japan Continued from page 1

scientists are working on cold fusion, but the number probably exceeds 1,000, and the research probably costs more than \$100 million per year. This is a modest, low key effort by the standards of industry. \$100 million is, after all, about what it would cost to build one computer chip factory, or to redesign an automobile. Most research is conducted in secret by major corporations like Toyota, Hitachi and Mitsubishi. It is very difficult for anyone to know how many of these companies are seriously working on the problem, and how much they are spending. Even the head of the government program, Dr. Ikegami, can only hazard a guess that "20 companies are seriously working on cold fusion."

In contrast, the government's role in cold fusion is open. It is a matter of public record. Research is conducted by the Ministry of Education, in University and in National Laboratories. All details of this research is completely open to the public. Anyone can contact the researchers, and they usually mail the latest papers and data. Any scientist or scientifically minded member of the general public can make arrangements to visit the laboratories and see work in progress. In 1991, the Ministry of International Trade and Industry (MITI) formed an industry consortium and launched a program to develop cold fusion into a practical form of energy. Information about this effort is harder to come by, because private industry does not generally reveal many details about ongoing research and development programs in any field.

Boom & Bust - A Gradual Rebirth

The early history of cold fusion in Japan followed the pattern seen in Europe and the United States. There was an initial boom after the news came out. Worldwide, several dozen experienced electrochemists got positive results within a few weeks of the announcement, and immediately went to work on the problem seriously, in places like Los Alamos, Texas A&M, and Stanford University. In dozens of other laboratories however, inexperienced physicists and chemists tried the experiment in a desultory fashion, without knowing enough about electrochemistry or the conditions necessary for success. After a few months, the vast majority of these people gave up in disgust. While this brouhaha ensued in the U.S., in Japan events unfolded much more slowly, in a far different fashion. First, the boom was not as great as it was in the Europe and America. Second, there was never any serious backlash. The majority of scientists, particularly physicists, did not believe that cold fusion could exist, and they did not attempt to replicate it in 1989. To this day, many of them remain skeptical. Hideo Ikegami, at the National Institute for (hot) Fusion Science reacted in a typical way. The Ministry of Education suggested to him that he shift a few percent of the hot fusion budget into cold fusion, but he refused, because he thought that cold fusion was a mistake. Later, he changed his mind, and now he has become one of Japan's leading cold fusion scientists.

In Japan, a small number of relatively experienced electrochemists and nuclear chemists began long term experiments. Many of them were

still calibrating and getting ready to begin the actually experimental work in the weeks following the (March 1989) announcement, while the American teams were quitting in droves. In June 1989, the Japanese edition of Scientific American took a serious look at cold fusion in a "special issue article" titled Room Temperature Fusion, Basic Knowledge and Actual Experimental Conditions. By the autumn of 1989, positive results were reported by the University of Osaka and the University of Hokkaido. A short article about the Osaka work appeared in a U.S. AP wire, but after this, for two years, to our knowledge nothing about the work in Japan appeared in the U.S. press, and matter, very little was published in Japan. In 1990, the Tokyo Univ. of Agriculture and Technology group reported excess heat in the Bulletin of the Chemical Society of Japan, and again on the first day of Nagoya Conference in 1992. The Atomic Energy Research group attended the Conference, but did not contribute a paper. Other people reporting early positive results included: Tadahiko Mizuno, Hokkaido National University Akito Takahashi, Osaka National University Eichi Yamaguchi, NIT Corporation. The U.S. and Japanese paths diverged radically from this time on. The DoE ERAB report signalled the end of U.S. participation in the field except at EPRI/SRI and Utah's National Cold Fusion Institute. In the U.S. and in Italy, "open season" was declared on cold fusion researchers and on scientists who supported the idea of research in this area. Britain's Nature magazine and other leading scientific journals declared cold fusion a mistake, and then a lie, and then a fraud. A full scale "witch hunt" got underway in the U.S., as the Washington Post, the Scientific American, Nature, the New York Times and dozens of others joined in the savage, personal attacks against cold fusion scientists. In Japan a few dozen scientists buckled down for the long haul. Many Japanese corporate labs, and some

national laboratories, were growing discouraged by late 1989, and dropping out of the field, with a shrug of the shoulders. Four books about cold fusion were published in 1989 and 1990, which ranged from cautious to optimistic.

The first Japanese scientist to announce positive results was Dr. Tadahiko Mizuno of Hokkaido National University. He was well qualified, he has been working with metal hydrides throughout his 20 year career in electrochemistry. He set up an experiment in the sub-basement at the university, which is designed to reduce background noise from cosmic and local sources to a low level. It has concrete walls many meters thick, and a gigantic lead door about a meter thick. In the fall of 1989, Mizuno detected neutrons. In 1990, he moved on to tests for tritium. His first significant burst of heat came in 1991. Mizuno says he came under severe criticism, and ran into a wall of disbelief when he announced these findings at the Electrochemical Society meetings and elsewhere. He got very little funding for the work, and at one point, he was even forced to dig into his own pocket to finance research. On the other hand, the Department Chairman supported the idea of doing cold fusion research. Mizuno did not have an easy time, but compared to his U.S. colleagues, he had a lot to be thankful for. At least he was allowed to speak and present papers at Electrochemical Society. In the U.S., not only would he be shut out of all meetings and publications, but it is likely that his superiors would threaten to fire him because he got positive results!

Other Japanese researchers who have reported positive results have also spoken of the harsh criticism and opposition they encounter. The Japanese edition of Playboy magazine

continued on page 9



The BEST Editorial Cartoon of the Quarter
from The Boston Herald (9/11/93)

We wish our Vice President well in his uphill quest against incompetence, redundancy, and all the non-responsive and non-optimal parts of the System.

THE DEVELOPMENT OF SPACE CONTINUED FROM PAGE 5

system, is a far better measure. Only the rocket burn method needs fuel by taking the sum of all velocity changes from burns of rocket engines in a mission we arrive at a number either called a Delta-V total or Velocity budget. This number is directly proportional to the fuel needed in the mission, and best short-hand representation of the expense of sending a spacecraft to a particular destination. An early published example of this type of analysis was done by Arthur C. Clarke in. He computed "velocity budgets" from the Earth's surface to the Earth's moon, Mars and Venus. In the past few years many papers have extended this analysis to the point where it is possible to assemble a good picture of the economic geography of the inner solar system and conceptualize a strategy for the movement of mankind out from Low Earth Orbit (LEO). Theoretical models suggest, Phobos is between 4-20 percent water by weight with nitrogen and carbon present. The only data we have at present is the low albedo measured by Viking which indicates a composition similar to C1 or C2 carbonaceous chondrite meteorites, rich in volatiles.

III. ASTEROID RESEARCH Another space enterprise that could emerge as a major industry in the 21st century is asteroid mining. There are only a few minerals valuable enough to justify extra terrestrial mining expeditions for an earth market. The platinum group of metals, platinum, palladium, osmium, iridium, ruthenium, and rhodium, range in value from about \$400 a troy ounce to \$1250 a troy ounce (for rhodium). They are key in many important industrial processes, particularly as catalysts. Ninety-four percent of the supply on Earth is controlled by the USSR and the Republic of South Africa. Rhodium, key to pollution control technology, is 3 times the price of gold and inflating rapidly. A shuttle load of rhodium (50,000 lbs) would be worth \$750,000,000 US at current world prices. Most of the earth's platinum metals are deep in the planet's iron nickel core; Asteroids, particularly the iron nickel type, which comprise perhaps 10% main belt asteroids, are thought to be richer in the platinum group metals than the best ores from the Earth's crust. Current research on prehistoric Earth impacts by extraterrestrial objects indicates that a high percentage of platinum metals, particularly iridium and osmium, is one of the major signatures of comets/asteroids that caused major extinctions in the geological past. There is a significant chance that even Phobos and Deimos (thought to be captured asteroids) may have economically viable concentrations of strategic metals. Speculation about Phobos and Deimos has centered on their suspected richness in volatiles for fuel and life support. The moons of Mars, Phobos and Deimos, may be natural sources of fuel, volatiles and bulk material for a LEO end-market. They are also good bases for the exploration of Mars.

Response Continued from page 6

about stages three or four using H_2O in place of D_2O . As has already been pointed out above he has failed to comment on the extensive discussion in our paper of a "blank experiment". Admittedly, the evidence was restricted to stages 1 and 2 of his own classification but a reference to an independent review of our own work [10] will show him and interested readers that such cells stay in thermal balance to at least 90 degrees C (we note that Douglas Morrison was present at the Second Annual Conference on Cold Fusion). We find statements of the kind made by Douglas Morrison distasteful. Have scientists now abandoned the notion of verifying their facts before rushing into print?

In the last paragraph of this section Douglas Morrison finally "boxes himself into a corner": having set up an unlikely and unworkable scenario he finds that this cannot explain Stage 5 of the experiment. In the normal course of events this should have led him to: (i) enquire of us whether the particular experiment is typical of such cells; (ii) to revise his own scenario. Instead, he implies that our experiment is incorrect, a view which he apparently shares with Tom Droege [11]. However, an experimental observation is just that: an experimental observation. The fact that cells containing palladium and palladium alloy cathodes polarized in D_2O solutions stay at high temperatures after they have been driven to such extremes of excess enthalpy generation does not present us with any difficulties. It is certainly possible to choose conditions which also lead to "boiling to dryness" in "blank cells" but such cells cool down immediately after such "boiling to dryness". If there are any difficulties in our observations, then these are surely in the province of those seeking explanations in terms of "Chemical Effects" for "Cold Fusion". It is certainly true that the heat transfer coefficient for cells filled with gas (N_2) stay close to those for cells filled with 0.1M LiOD (this is not surprising because the main thermal impedance is across the vacuum gap of the Dewar-type cells). The "dry cell" must therefore have generated ~120kJ during the period at which it remained at high temperature (or ~3MJcm³ or 26MJ/mol Pd⁻¹). We refrained from discussing this stage of the experiments because the cells and procedures we have used are not well suited for making quantitative measurements in this region. Inevitably, therefore, interpretations are speculative. There is no doubt, however, that Stage 5 is probably the most interesting part of the experiments in that it points towards new systems which merit investigation. Suffice it to say that energies in the range observed are not within the realm of any chemical explanations.

We do, however, feel that it is justified to conclude with a further comment at this point in time. Aficionados of the field of "Hot Fusion" will realize that there is a large release of excess energy during Stage 5 at zero energy input. The system is therefore operating under conditions which are described as "Ignition" in "Hot Fusion". It appears to us therefore that these types of systems not only "merit investigation" (as we have stated in the last paragraph) but, more correctly, "merit frantic investigation".

Morrison's "Conclusions" and General Comments

In his section entitled "Conclusions", Douglas Morrison shows yet again that he does not understand the nature of our experimental techniques, procedures and

methods of data evaluation (or, perhaps, that he chooses to misunderstand these?). Furthermore, he fails to appreciate that some of his own recommendations regarding the experiment design would effectively preclude the observation of high levels of excess enthalpy.

We illustrate these shortcomings with a number of examples:

(i) Douglas Morrison asserts that accurate calorimetry requires the use of three thermal impedances in series and that we do not follow this practice. In point of fact we do have three impedances in series: from the room housing the experiments to a heat sink (with two independent controllers to thermostat the room itself); from the thermostat tanks to the room (and, for good measure, from the thermostat tanks to further thermostatically controlled sinks); finally, from the cells to the thermostat tanks. In this way, we are able to maintain 64 experiments at reasonable cost at any one time (typically two separate five-factor experiments).

(ii) It is naturally essential to measure the heat flow at one of these thermal impedances and we follow the normal convention of doing this at the innermost surface (we could hardly do otherwise with our particular experiment design!). In our calorimeters, this thermal impedance is the vacuum gap of the Dewar vessels which ensures high stability of the heat transfer coefficients. The silencing of the top section of the Dewars (see Fig 2 of our paper [2]) further ensures that the heat transfer coefficients are virtually independent of the level of electrolyte in the cells.

(iii) Douglas Morrison suggests that we should use isothermal calorimetry and that, in some magical fashion, isothermal calorimeters do not require calibration. We do not understand: how he can entertain such a notion? All calorimeters require calibration and this is normally done by using an electrical resistive heater (following the practice introduced by Joule himself). Needless to say, we use the same method. We observe that in many types of calorimeter, the nature of the correction terms are "hidden" by the method of calibration. Of course, we could follow the self-same practice but we choose to allow for some of these terms explicitly. For example, we allow for the enthalpy of evaporation of the D_2O . We do this because we are interested in the operation of the systems under extreme conditions (including "boiling") where solvent evaporation becomes the dominant form of heat transfer (it would not be sensible to include the dominant term into a correction).

(iv) There is, however, one important aspect which is related to (iii) i.e. the need to calibrate the calorimeters. If one chooses to measure the lower bound of the heat transfer coefficient (as we have done in part of the paper published recently in this journal [2]) then there is no need to carry out any calibrations nor to make corrections. It is then quite sufficient to investigate the time dependence of this lower bound heat transfer coefficient in order to show that there is a generation of excess enthalpy for the Pd- D_2O system whereas there is no such generation for appropriate blanks (e.g. Pt- D_2O or Pd- H_2O). Alternatively, one can use the maximum value of the lower bound heat transfer coefficient to give lower bound values of the rates of excess enthalpy generation. It appears to us that Douglas Morrison has failed to understand this point as he continuously asserts that our demonstrations of excess enthalpy generation are dependent on calibrations and corrections.

(v) Further with regard to (iii) it appears to us that Douglas Morrison believes that a "null method" (as used in isothermal calorimeters) is inherently more accurate than say the

Continued on Page 10

Congressional Hearing

CONTINUED FROM PAGE 4:

for future generations, for those who will live in the twenty-first century."

The two key action items that CFRA emphasized were: (1) To allow a \$300K study by the National Academy of Sciences (NAS) of "Alternative Fusion Energy" and (2) Hold hearings on cold fusion in the relevant House and Senate Committees. These are the goals which CFRA feels are most immediately attainable.

Ed. The Staff of COLD FUSION TIMES acknowledges the courage of those Congresspeople who are looking into this field, and what happened to it in mid- and late-'89. The TRIGGER poll (page 9) offers a glimpse into the divergent points of view over cold fusion elsewhere (in Japan).

No Portion of this journal may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means, including electronic, print, photocopy, mechanical, recording or otherwise, without the prior permission of
JET Technology
c/o COLD FUSION TIMES

EPRI Leadership in Science and Technology

Fourth International Conference (ICCF-4) Four years of intensive investigation have uncovered a wide variety of unexpected phenomena occurring in condensed matter under ambient conditions. The phenomena in question and their associated disciplines include excess power generation, nuclear product formation, electrochemical studies of deuterated metal systems, solid-state physics of metal matrices, coherent processes, behavior of gas-metal systems, and improved precision calorimetric techniques.

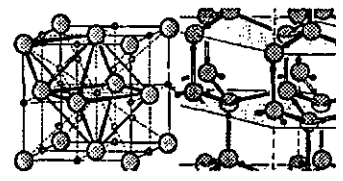
Information Linda Nelson, EPRI,
(415) 855-2127

The MIT Entrepreneurs Club

aka: "The E-Club"
weekly meetings, year-round
every Tuesday at 6pm & 7pm
66-144 & 66-148 at MIT

"No-Tech to High-Tech Start-Up Help"

Information: v-mail & info-tree
(617) 253-2000
e-newsletter subscriptions:
e-club_request@athena.mit.edu
other questions contact
E-Club Co-Founder/Co-Director:
Richard Shyduroff, tele: (617)
253-2000, fax: 253-8000
Internet:



COLD FUSION TIMES

Cold Fusion in Japan

Continued from page 7

interviewed Dr. Takahashi. The interviewer, Teruaki Fukami, suggested that before the Japanese physics establishment accepted cold fusion, Takahashi and others would have to: "...take part in their physics conferences, and publish in their journals. I see you have not done that yet. Can you tell us why, not?"

Takahashi, a modest, easy going man, responded characteristically: "Well, I am not the confrontational type; I do not look for a quick, decisive outcome. I don't want that. At those physics conference there would be many of these so called renowned and famous physicists, and most of them are confirmed skeptics, who don't believe cold fusion exists. I cannot see myself charging in there with a bamboo spear, like Don Quixote attacking a windmill, and getting into an academic debate. If we are going to have an academic battle, I would prefer to prepare the forces gradually, and gradually undermine them, surround them; take the castle one moat at time, as the expression goes. The critical thing is to build up the idea over time, until the mood swings in our direction, and they invite me in to do a lecture."

Later in the interview, he said: "...cold fusion has not become acceptable to the larger world of science yet, by a long shot; it will take a lot more work. It won't happen until all those famous professors at physics conferences - the people we were talking about before - accept that cold fusion is real... As for me, why, during the Nagoya meeting, me and my theory might be drawn, quartered, and thrown into the trash."

In his book *Capturing the Sun in A Test Tube*, Yamaguchi describes the intense barrage of attacks he came under after announcing at the Nagoya Conference that he had detected commensurate amounts of helium-4. He felt overwhelmed by the hysterical, jealousy, hate and malice, and he felt appalled that scientist would act in such unscrupulous ways. But all that is no longer open to discussion. Cold fusion experiments and replication left those levels of doubt a long time ago, and entered a more concrete stage of development. Anyone who still says, "such nonsense, it can't be", is simply not looking at reality." In December 1991, Bungeishunju published a two page photo-essay, titled "Finally off an running: research on 'the ultimate dream energy'".

The Attitudes of Decision Makers - The Trigger Poll Many Americans want to know what the middle aged "decision makers" in the Japanese scientific establishment think of cold fusion. Do scientists in industry, in the universities and in the national laboratories think cold fusion is real? Do they think it will become a practical source of energy? In most countries, we would be forced to speculate on such questions, but fortunately, Japanese publishers are very fond of public opinion research and polling, so we have detailed answers. The Nikkei publishing company's *Trigger* magazine did a high quality public opinion survey, which was featured on the front cover of the June, 1993 edition, Vol. 12, No. 6. *Trigger* sent out 300 questionnaires to leading scientists at universities, industrial corporations, and national laboratories. They got back 180 responses, a 63% response rate, which is a staggering high response rate, even for Japan. The respondents represent the decision making class of Japan: the largest age group represented is the 55 - 59 year olds, 24% of the total.

Not only was the response rate high, but the editors of *Trigger* were flabbergasted to find that "over 100" respondents included extensive additional comments, averaging about 200 words apiece, but some including a whole page or two of comments.

The editors printed four pages of excerpts from the comments. Here are a few of the questions and number of responses.

Regarding cold fusion, "Drs. Pons and Fleischmann, and Dr. Jones, announced the work. Yet replication experiments are still in progress, and the work is still controversial. Why do you think it has taken this amount of time?"

- A. Because the effect does not exist, and attempts to replicate it are a waste of time. 27
- B. Because the replications attempts have been low quality scientific work. 23
- C. Because it is a new phenomenon, and it is natural that replication should take a long time. 127
- D. Don't know 3

"In 1989, the discovery of Pons and Fleischmann was first announced by the London interest in the field. Do you think this *Financial Times*, which triggered a world-wide fever is an appropriate forum for the announcement of a scientific discovery?"

- A. It was appropriate 39
- B. It was inappropriate 106
- C. Don't know 35

"The Ministry of International Trade and Industry (MITI) is now funding cold fusion research. How do you feel about this?"

- A. The nation should support cold fusion research. 60
- B. The nation need not support this research. 21
- C. If cold fusion turns out to be real, then a large-scale national project will be in order, but support at present is premature. 85
- D. Don't know. 12

"What do you think of the future prospects for cold fusion?"

- A. I think it will become a practical, desirable form of energy 29
- B. I think the prospects for practical use of cold fusion are poor 95
- C. Don't know 52

"How would you characterize media coverage of cold fusion?"

- A. There have been enough reports 16
- B. There has not been enough coverage 42
- C. There have been many reports, but the quality of these reports is low 111
- D. Don't know 8

Ed. note: These short excerpts are from **COLD FUSION ALMANAC** (in preparation - see page 10).

COLD FUSION TIMES

invites the reader to take a closer look at the **TRIGGER** Survey of FUSION

In What Time Period will Hot Fusion be made Practical, if it can be?

- mid 21st Century
- Later 21st Cent.
- ▨ During 22nd Cent.
- ⊠ Don't Know

Who Took the TRIGGER poll?

- Corporations
- Universities
- ▨ National Labs
- ⊠ Other

HOT FUSION - WHAT ARE THE FUTURE PROSPECTS

- No hope to be practical
- Difficult, continue to make practical
- ▨ Can be made practical
- ⊠ Don't know

COLD FUSION REPLICATION EXPERIMENTS ARE STILL IN PROGRESS AND THE WORK REMAINS CONTROVERSIAL. WHY DO YOU THINK IT HAS TAKEN THIS AMOUNT OF TIME?

- It does not exist
- Replication attempts are poor quality work.
- ▨ New phenomena naturally take a long time to replicate
- ⊠ Don't know

FOR THE WORLD (and more information)

10

isoperibolic calorimetry which we favour. While it is certainly believed that "null" methods in the Physical Sciences can be made to be more accurate than direct measurements (e.g. when a voltage difference is detected as in bridge circuits: however, note that even here the advent of "ramp" methods makes this assumption questionable): this advantage disappears when it is necessary to transduce the primary signal. In that case the accuracy of all the methods is determined by the measurement accuracy (here of the temperature) quite irrespective of which particular technique is used. In point of fact and with particular reference to the supposed advantages of isothermal versus isoperibolic calorimetry, we note that in the former the large thermal mass of the calorimeter appears across the input of the feedback regulator. The broadband noise performance of the system is therefore poor; attempts to improve the performance by integrating over long times drive the electronics into 1/f noise and, needless to say, the frequency response of the system is degraded. (see also (vii) below)

(vi) with regard to implementing measurements with isothermal calorimeters, Douglas Morrison recommends the use of internal catalytic recombiners (so that the enthalpy input to the system is just E_{cell} * I rather than $(E_{\text{cell}} - E_{\text{electrochemical, cell}}) * I$ as in our "open" calorimeters. We find it interesting that Douglas Morrison will now countenance the introduction of intense local "hot spots" on the recombiners (what is more in the gas phase!) whereas in the earlier parts of his critique he objects to the possible creation of microscopic "hot spots" on the electrode surfaces in contact with the solution.

We consider this criticism from Douglas Morrison to be invalid and inapplicable. In the first place it is inapplicable because the term $E_{\text{electrochemical, cell}}$ * I (which we require in our analysis) is known with high precision (it is determined by the enthalpy of formation of D_2O from D_2 and $1/2 O_2$). In the second place it is inapplicable because the term itself is -0.77 Watt whereas we are measuring a total enthalpy output of ~ 170 Watts in the last stages of the experiment.

(vii) We observe here that if we had followed the advice to use isothermal calorimetry for the main part of our work, then we would have been unable to take advantage of the "positive feedback" to drive the system into regions of high excess enthalpy generation (perhaps, stated more exactly, we would not have found that there is such positive feedback). The fact that there is such feedback was pointed out by Michael McKubre at the Third Annual Conference of Cold Fusion and strongly endorsed by one of us (M.F.). As this issue had then been raised in public, we have felt free to comment on this point in our papers (although we have previously drawn attention to this fact in private discussions). We note that Douglas Morrison was present at the Third Annual Conference on Cold Fusion.

(viii) While it is certainly true that the calorimetric methods need to be evolved, we do not believe that an emphasis on isothermal calorimetry will be useful. For example, we can identify three major requirements at the present time: a) the design of calorimeters which allow charging of the electrodes at low thermal inputs and temperatures below 50 degrees C followed by operation at high thermal outputs and temperatures above 100 degrees C, b) the design of calorimeters which allow the exploration of Stage 5 of the experiments, c) the design of calorimeters having a wide frequency response in order to explore the transfer functions of the systems. We note that c) will in itself lead to calorimeters having an accuracy which could hardly be rivalled by other methods.

(ix) Douglas Morrison's critique implies that we have never used calorimetric techniques other than that described in our recent paper [2]. Needless to say, this assertion is incorrect. It is true, however, that we have never found a technique which is more satisfactory than the isoperibolic method which we have described. It is also true that this is the only method which we have found so far which can be implemented within our resources for the number of experiments which we consider to be necessary. In our approach we have chosen to achieve accuracy by using software; others may prefer to use hardware. The question as to which is the wiser choice is difficult to answer: it is a dilemma which has to be faced frequently in modern experimental science. We observe also that Douglas Morrison regards complicated instrumentation (three feedback regulators working in series) as being "simple" where he regards data analysis as being complicated. Douglas Morrison also asserts that we have never used more than one thermistor in our experimentation and he raises this issue in connection with measurements on cells driven to boiling. Needless to say, this assertion is also incorrect. However, further to this remark it is necessary for us to point out that one does not need any temperature measurement in order to determine the rate of boiling of a liquid?

(x) Douglas Morrison evidently has difficulties with our application of non-linear regression methods to fit the integrals of the differential equations to the experimental data. Indeed he has such an idee fixe regarding this point that he maintains that we used this method in our recent paper [2]; we did not do so (see also 'stage 3 calculations' above). However, we note that we find his attitude to the Levenberg-Marquardt algorithm hard to understand. It is one of the most powerful, easily implemented "canned software" methods for problems of this kind. A classic text for applications of this algorithm [12] has been praised by most prominent physics journals and magazines.

(xi) Douglas Morrison's account contains numerous misleading comments and descriptions. For example, he refers to our calorimeters as "small transparent test tubes". It is hard for us to understand why he chooses to make such misleading statements. In this particular case he

Continued page 11

We want to thank the many readers, contributors, supporters, helpers, consultants, and friends who have continued to help bring the **COLD FUSION TIMES** to fruition.



The World
A Public Information Utility
Software Tool & Die
1330 Beacon Street, Suite 215
Brookline, Massachusetts
02146
office@world.std.com
617-739-0202

NuTECH is actively seeking interested individuals in the field of cold fusion. Send your resume', or background, or brief summary of your interests to **NuTECH c/o COLD FUSION TIMES** (Dept. N3-1)

The Memory Location



11th Washington Street, Wellesley, MA 02158
(617) 237-6846 Voice • (617) 237-9525 Fax
The Store for Amiga Explorers



The Cold Fusion Newsletter
One (1) year (bimonthly) \$180.00
PO Box 60642
Palo Alto, CA 94306

A VERY SPECIAL OPPORTUNITY

Want to Learn About COLD FUSION?
Special--- only \$89 (*)

- 1) "Fire From Ice" by Dr. Eugene Mallove (Wiley)
 - 2) "COLD FUSION IMPACT" by Hal Fox
 - 3) A 1 year subscription to COLD FUSION TIMES write to COLD FUSION TIMES, BOX C-3 (* add \$15.00 [US currency only please] for orders outside of the United States of America)
- Offer ends Dec. 21, 1993 and is void where not allowed

FUSION FACTS

P.O. Box 58639
Salt Lake City, UT 84158
(801) 583-6232

12 issues (1 year) \$345



Minnesota Cold Fusion Alliance

#2 223 Ridgewood
Minneapolis, MN 55403

JET Technology

Consulting Services for
Material and Corrosion
Science and Engineering
1 (617) 239-8383

BOOKS

AVAILABLE ON THE SUBJECT
OF **COLD FUSION**

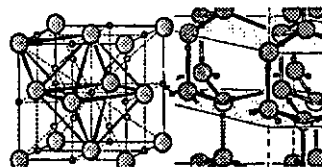
COLD FUSION IMPACT \$24.95
FIRE FROM ICE \$22.95
COLD FUSION ALMANAC (~\$60 in preparation)

There is an extra postage and handling charge of \$6.00 [US currency only please] for orders outside of the United States.

COLD FUSION TIMES

11

from Valbonne, France --Response from page 10



could equally well have said "glass Dewar vessels silvered in their top portion" (which is accurate) rather than "small transparent test tubes" (which is not). Alternatively, if he did not wish to provide an accurate description, he could simply have referred readers to Fig 2 of our paper [2]. This type of misrepresentation is a non-trivial matter. We have never used calorimeters made of test-tubes since we do not believe that such devices can be made to function satisfactorily.

(xii) As a further example of Douglas Morrison's inaccurate reporting, we quote his last paragraph in full: "It is interesting to note that the Fleischmann and Pons paper compares their claimed power production with that from nuclear reactions in a nuclear reactor and this is in line with their dramatic claims (9) that **"SIMPLE EXPERIMENT" RESULTS IN SUSTAINED N-FUSION AT ROOM TEMPERATURE FOR THE FIRST TIME: breakthrough process has potential to provide inexhaustible source of energy.** It may be noted that the present paper does not mention "Cold Fusion" nor indeed consider a possible nuclear source for the excess heat claimed."

Douglas Morrison's reference (9) reads: Press release, University of Utah, 23 March 1989. With regard to this paragraph we note that: a) our claim that the phenomena cannot be explained by chemical or conventional physical processes is based on the energy produced in the various stages and not the power output, b) the dramatic claim he refers to was made by the Press Office of the University of Utah and not by us, c) we did not coin the term "Cold Fusion" and have avoided using this term except in those instances where we refer to other research workers who have described the system in this way. Indeed, if readers refer to our paper presented to the Third International Conference on Cold Fusion [13] (which contains further information about some of the experiments described in [2]), they will find that we have not used the term there. Indeed, we remain as convinced as ever that the excess energy produced cannot be explained in terms of the conventional reaction paths of "Hot Fusion", and d) it has been widely stated that the editor of this journal "did not allow us to use the term Cold Fusion". This is not true: he did not forbid us from using this term as we never did use it (see also [13]).

(xiii) In his section "Conclusions", Douglas Morrison

makes the following summary of his opinion of our paper: "The experiment and some of the calculations have been described as 'simple'. This is incorrect - the process involving chaotic motion, is complex and may appear simple by incorrectly ignoring important factors. It would have been better to describe the experiments as 'poor' rather than 'simple'. We urge the readers of this journal to consult the original text [2] and to read Douglas Morrison's critique [1] in the context of the present reply. They may well then come to the conclusion that our approach did after all merit the description 'simple' but that the epithet 'poor' should be attached to Douglas Morrison's critique."

Our own conclusions We welcome the fact that Douglas Morrison has decided to publish his criticisms of our work in the conventional scientific literature rather than relying on the electronic mail, comments to the press and popular talks; we urge his many correspondents to follow his example. Following this traditional pattern of publication will ensure that their comments are properly recorded for future use and that the rights of scientific referees will not be abrogated. Furthermore, it is our view that a return to this traditional pattern of communication will in due course eliminate the illogical and hysterical remarks which have been so evident in the messages on the electronic bulletins and in the scientific tabloid press. If this proves to be the case, we may yet be able to return to a reasoned discussion of new research. Indeed, critics may decide that the proper course of inquiry is to address a personal letter to authors of papers in the first place to seek clarification of inadequately explained sections of publications. Apart from the general description of stages 1-5, we find that the comments made by Douglas Morrison are either irrelevant or inaccurate or both.

References

- [1] Douglas Morrison, Critique published on Internet
- [2] M.Fleischmann and S. Pons, Phys. Lett. A 176 (1993) 1
- [3] to be published
- [4] M.Fleischmann, S.Pons, M.W.Anderson, L.J. Li, and M. Hawkins, J. Electroanal. Chem. 287 (1990) 293.
- [5] R.H. Wilson, J.W. Bray, P.G. Kosky, H.B. Vakil, and F.G. Will, J. Electroanal. Chem. 332 (1992) 1
- [6] M.Fleischmann and S.Pons, J. Electroanal. Chem. 332 (1992) 33
- [7] S. Pons and M.Fleischmann in : Final Report to the Utah State Energy Advisory Council, June 1991.
- [8] G. Kreysa, G. Marx, and W.Plieth, J. Electroanal. Chem. 268 (1989) 659
- [9] M. Fleischmann and S. Pons, J. Electroanal. Chem. 261 (1989) 301
- [10] W.Hansen, Report to the Utah State Fusion Energy Council on the Analysis of Selected Pons-Fleischmann Calorimetric Data, in: "The Science of Cold Fusion": Proc. Second Annual Conf. on Cold Fusion, Como, Italy, 29 June-4 July 1991, eds T. Bressani, E. del Giudice and G. Preparata, Vol 33 of the Conference Proceedings of the Italian Physical Society (Bologna, 1992) p491; ISBN-887794-045-X
- [11] T. Droege: private communication to Douglas Morrison.
- [12] W.H. Press, B.P. Flannery, S.A. Teukolsky, and W.T. Vetterling, "Numerical Recipes", Cambridge University Press, Cambridge, 1989.
- [13] M.Fleischmann and S. Pons "Frontiers of Cold Fusion" ed. H. Ikegami, Universal Academy Press Inc., Tokyo, 1993, p47; ISBN 4-946-443-12-6

(*) Ed. - We apologize to our readers for the shortened "What's Happenin'" column with "From News in the Internet" but we are fortunate to have here the up-to-date reply of the two inventors of electrolytic loading of an isotopic material into a material. Although they may be modest in their referral to COLD FUSION and the reactions which take place therein, we are grateful both for their discovery and their timely reporting of it.

WHAT'S HAPPENIN'

[Including News from the INTERNET (*)]

"... when we made tritium it was pretty spectacular, I mean when you hold the holy grail in your hand, it is hard to deny its existence."

[Dr. Edmund Storms, Los Alamos National Laboratory, CBC News]

"In Harwell's D2O Cell 3 there are more than ten time intervals where an unexplained power source or energy storage mechanism may be operating. *** Harwell Cells 1,2,3 and 4 were wired in series to a constant current source. *** Throughout these anomalous increases in temperature in Cell 3, Cell 4 behaves "normally", i.e., it suffers no unexplained pulses of energy. Our initial estimate of the power associated with these anomalous temperature increases is 100-200 mW."

[Melich, M.E., Hansen, W.N., "Some Lessons from 3 Years of Electrochemical Calorimetry", in ICCF-3 Frontiers of Cold Fusion", Universal Academy Press, Inc. (1993)]

Proposed Federal Actions to Regain U.S. Leadership in Cold Fusion

(based upon CFRA; see page 1)

I. Remove Obstacles

- * Allow NAS Study on Alternative Fusion
- * Cold Fusion Research Category for SBIR
- * Identify Reasons for Obstacles at Patent Office

II. Promote Involvement at National Level

- * Hold Hearings on Cold Fusion to:
 - Review New Developments
 - Examine International Aspects
 - Develop U.S. Initiatives

III. Fund Research

- * Existing at Federal Labs
- * University Research in Cold Fusion
- * Innovative Research

from Toronto, Canada

The Secret Life of Cold Fusion

Continued from Page 12

institutions that have reported positive results. The original experiment has been refined and improved by Pons and Fleischmann in a state-of-the-art laboratory in France funded by Japan's giant Toyota corporation. Thompson interviewed Dr. Pons in his new laboratory about the continued work on cold fusion. He also reported on scientists from research institutes in Great Britain, Russia, Japan, and the United States about the Pons/Fleischmann experiment and alternative methods of creating cold fusion. And he reported from those who discount the entire idea of cold fusion -- a majority that is slowly eroding as more and more scientists become convinced that something extraordinary did happen in Utah four years ago. [CBC Prime Time News was a production of CBC News.]

COLD FUSION TIMES

Publisher and Editor: Mitchell Swartz
P.O. Box 81135, Wellesley Hills, Massachusetts 02181
(617) 239-8383 Internet address: mica@world.std.com

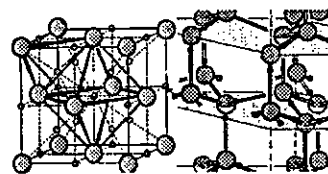
Contributors and Consultants for this issue: Stephen Baer, Eugene Mallove, Malcolm Partridge, Jed Rothwell, Dana Rotegard, Julius Mariasis, Leonard Kessel, Richard Goldbaum, Heidi LaFleche, Jeffery Tolleason, Don Towne, Gayle Verner, John Voci, Jayne Yaffe

COLD FUSION TIMES is published quarterly. Annual subscription rate is \$50.00 per year. Single issues are \$16.00. There is an extra postage charge outside North America of \$12.00 per year; US currency only please. ©1993 JET Technology
All Rights Reserved

COLD FUSION TIMES

TO COLDLY GO WHERE NO ONE
HAS GONE BEFORE

© copyright 1993
JET Technology



VOLUME 1 NUMBER 3
FALL '93

The Secret Life of Cold Fusion A CBC Prime Time News Documentary

CBC Prime Time News presented **The Secret Life of Cold Fusion**, a half-hour investigative documentary on Thursday June 24 at 9 p.m. on CBC Television. Journalist Jerry Thompson and Producer Robin Christmas reported that research continues quietly in laboratories around the world on a scientific breakthrough that, if it proves to be true, could help solve the world's energy problems beyond the next millennium.

Four years ago near Salt Lake City, when Dr. Stanley Pons and Dr. Martin Fleischmann announced that they had been able to create cold fusion using a car battery and a test tube, they became overnight celebrities. Their theory was widely discounted by a skeptical scientific community almost as quickly. Since then, while cold fusion has disappeared from the headlines, a number of elite and powerful research labs and companies around the world have been conducting their own experiments, away from the glare of publicity. In researching this story, Prime Time News was contacted by twenty nine

Continued Page 11

Subscriptions &
Editorial information:
COLD FUSION TIMES
P.O. Box 81135, Wellesley
Hills, Massachusetts 02181.

**COLD FUSION
TIMES**

ISSN# 1072-2874

Published quarterly

Annual rate is \$50.00 per year. Single
issues are \$16.00.

There is an extra postage charge outside
North America of \$12.00 per year.
Offer Void where prohibited by law.

COLD FUSION TIMES
P.O. Box 81135
Wellesley Hills, Massachusetts
02181

TO: